Maharashtra Education Society's

Institute of Management and Career Courses (IMCC), Pune

AUTONOMOUS

IT13: ADBMS

Chapter 2:SQL

Practice Assignment Solution

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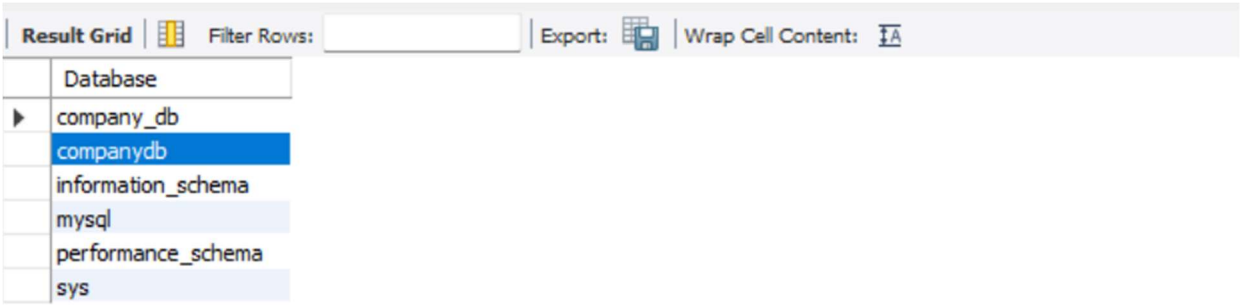
Write a query and paste a screenshot of your output below the question.

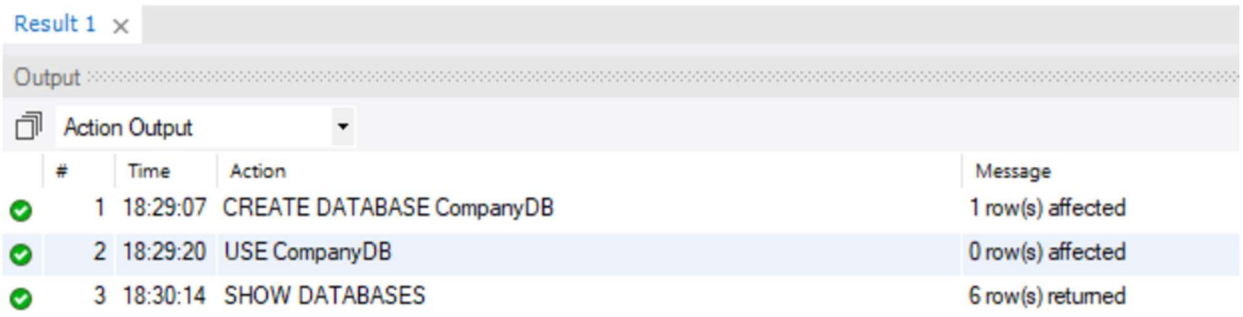
1. Create a database named CompanyDB.

   CREATE DATABASE CompanyDB;

   USE CompanyDB;

   SHOW DATABASES;





2. Create a table Employees with the following columns:

   Employees (EmpID (Integer, Primary Key), EmpName, Salary, JoinDate)

```
CREATE TABLE Employees (
    EmpID INT PRIMARY KEY,
EmpName VARCHAR(100),
Salary DECIMAL(12,2),
    JoinDate DATE,
Phone VARCHAR(13),
Age INT,
DeptID INT
    );
desc Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(100)	YES		NULL	
Salary	decimal(12,2)	YES		NULL	
JoinDate	date	YES		NULL	
Phone	varchar(13)	YES		NULL	
Age	int	YES		NULL	
DeptID	int	YES		NULL	

Result 2

Output

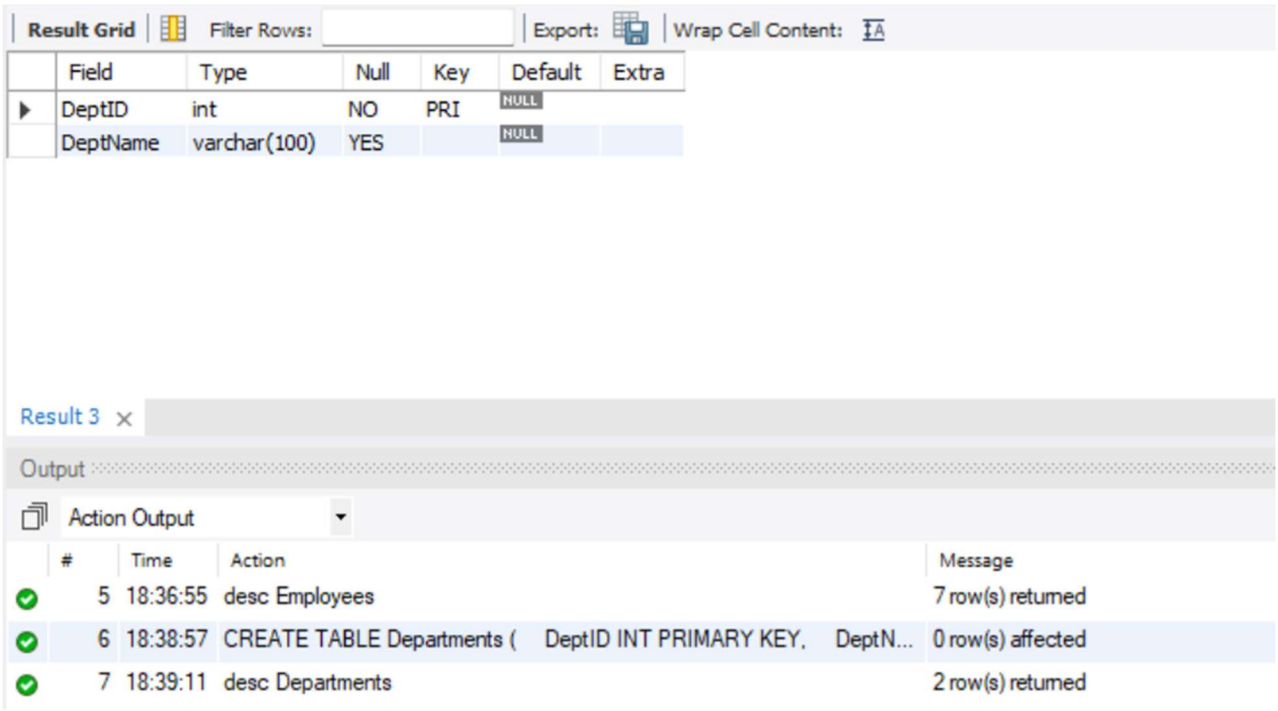
Action Output

#	Time	Action	Message
3	18:30:14	SHOW DATABASES	6 row(s) returned
4	18:34:35	CREATE TABLE Employees (EmpID INT PRIMARY KEY, EmpName ...	0 row(s) affected
5	18:36:55	desc Employees	7 row(s) returned

3. Create a table Departments with:

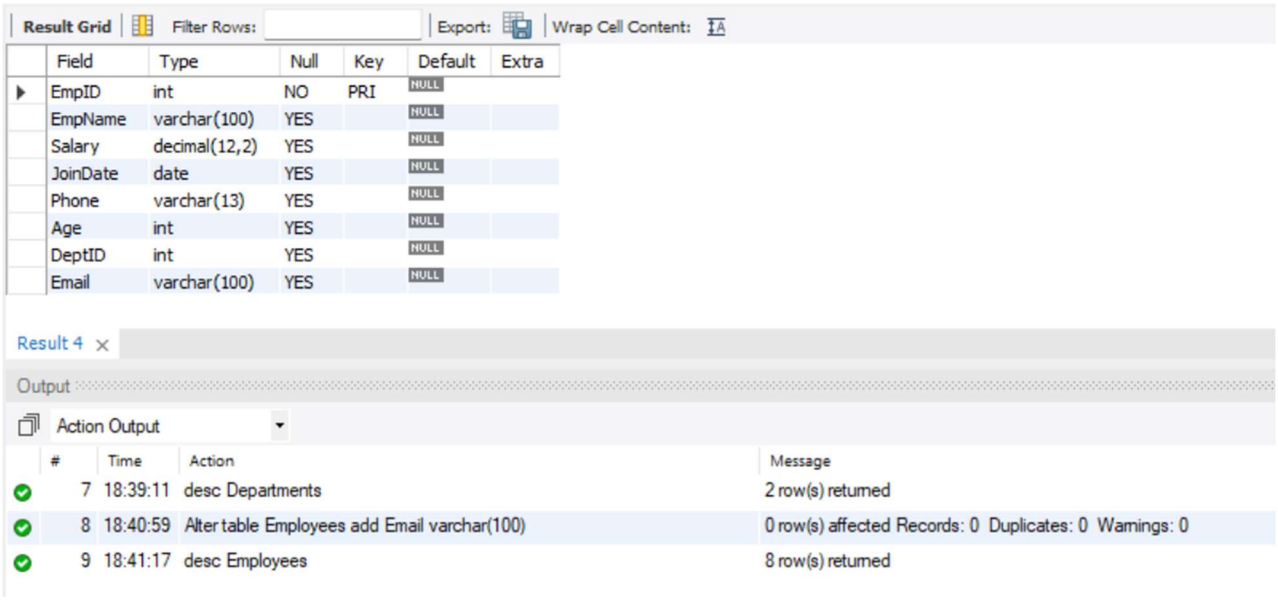
Departments (DeptID (Integer, Primary Key), DeptName)

```
CREATE TABLE Departments (
    DeptID INT PRIMARY KEY,
    DeptName VARCHAR(100)
        );
    desc Departments;
```



4. Add a column Email (VARCHAR(100)) to the Employees table.

```
Alter table Employees add Email varchar(100);
desc Employees;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(100)	YES		NULL	
Salary	decimal(12,2)	YES		NULL	
JoinDate	date	YES		NULL	
Phone	varchar(13)	YES		NULL	
Age	int	YES		NULL	
DeptID	int	YES		NULL	
Email	varchar(100)	YES		NULL	

Result 4

Output

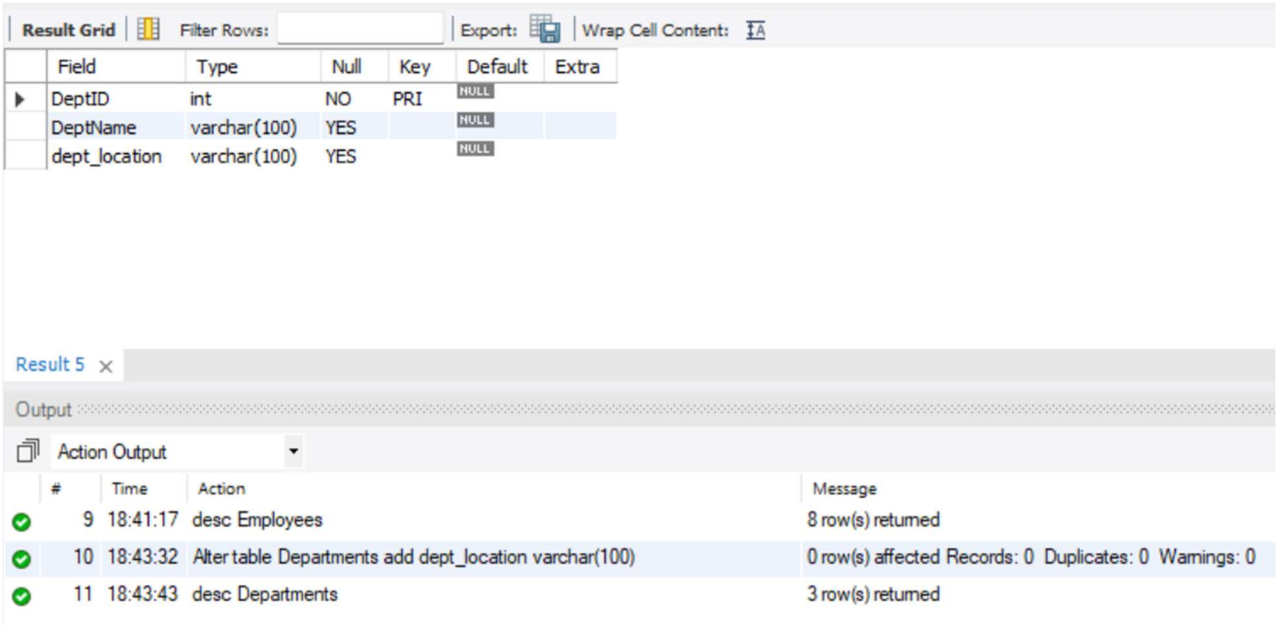
Action Output

#	Time	Action	Message
7	18:39:11	desc Departments	2 row(s) returned
8	18:40:59	Alter table Employees add Email varchar(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
9	18:41:17	desc Employees	8 row(s) returned

5. Add a column dept_location to the Departments table.

   Alter table Departments add dept_location varchar(100);

   desc Departments;



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
DeptID	int	NO	PRI	NULL	
DeptName	varchar(100)	YES		NULL	
dept_location	varchar(100)	YES		NULL	

Result 5

Output

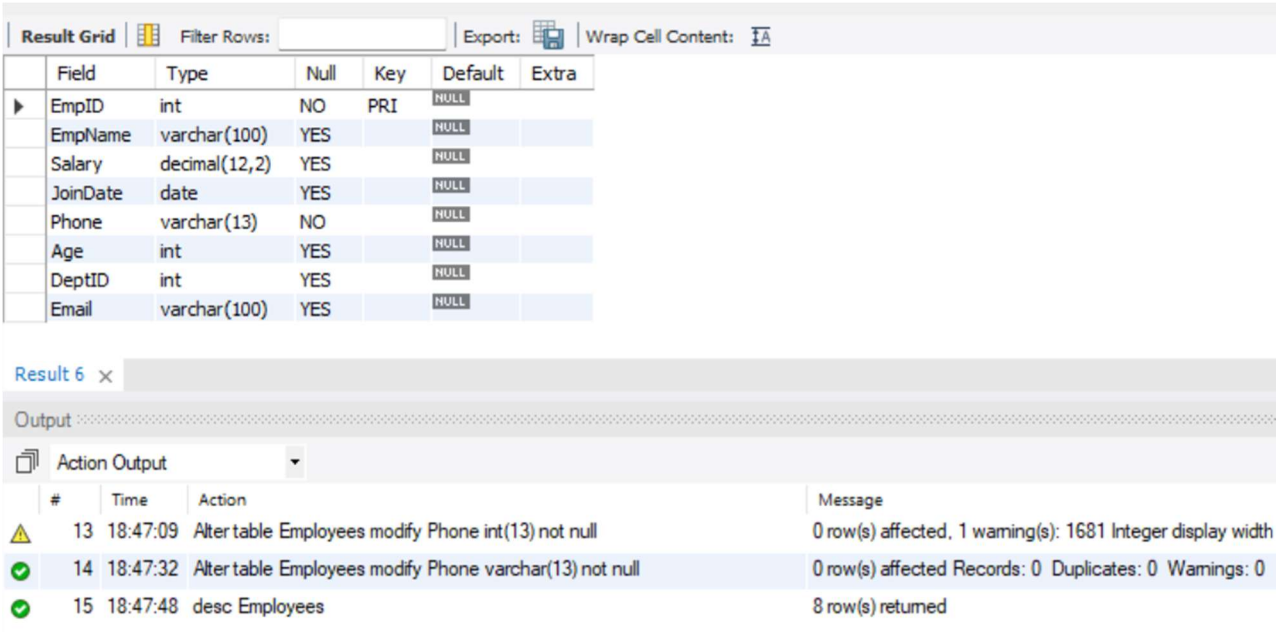
Action Output

#	Time	Action	Message
9	18:41:17	desc Employees	8 row(s) returned
10	18:43:32	Alter table Departments add dept_location varchar(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
11	18:43:43	desc Departments	3 row(s) returned

6. Add a NOT NULL Constraint to the Phone column

Alter table Employees modify Phone varchar(13) not null;

desc Employees;



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(100)	YES		NULL	
Salary	decimal(12,2)	YES		NULL	
JoinDate	date	YES		NULL	
Phone	varchar(13)	NO		NULL	
Age	int	YES		NULL	
DeptID	int	YES		NULL	
Email	varchar(100)	YES		NULL	

Result 6

Output

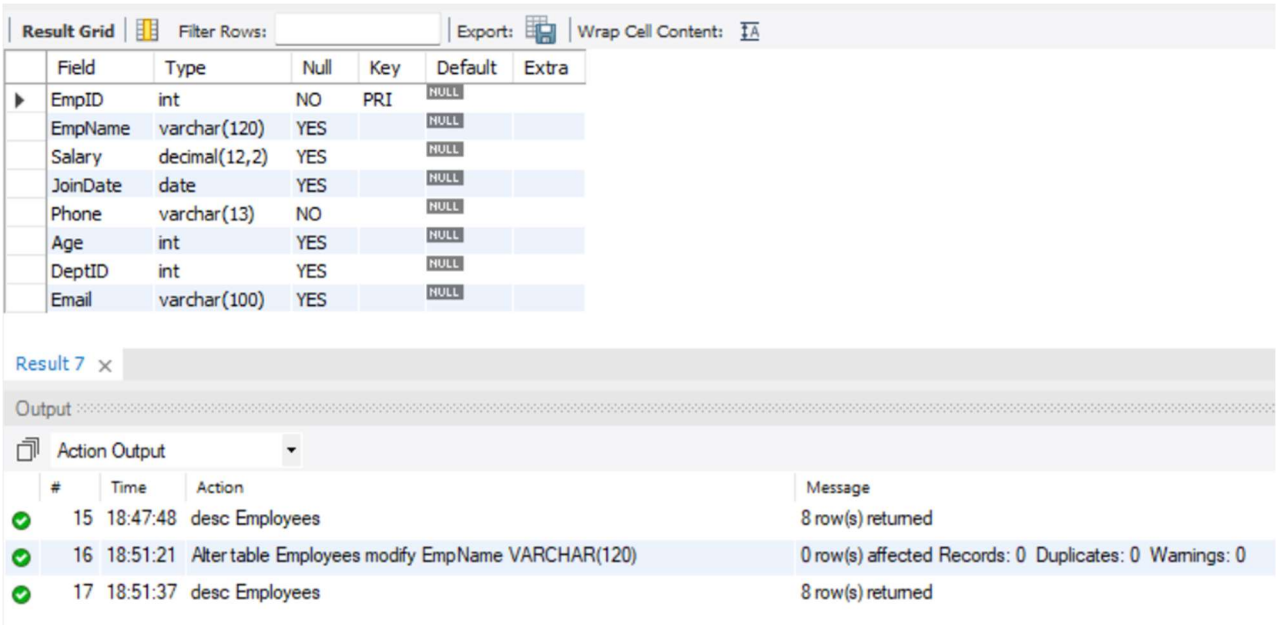
Action Output

#	Time	Action	Message
13	18:47:09	Alter table Employees modify Phone int(13) not null	0 row(s) affected, 1 warning(s): 1681 Integer display width
14	18:47:32	Alter table Employees modify Phone varchar(13) not null	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
15	18:47:48	desc Employees	8 row(s) returned

7. Change the size of the EmpName column from VARCHAR (50) to VARCHAR (100).

   Alter table Employees modify EmpName VARCHAR(120);

   desc Employees;



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(120)	YES		NULL	
Salary	decimal(12,2)	YES		NULL	
JoinDate	date	YES		NULL	
Phone	varchar(13)	NO		NULL	
Age	int	YES		NULL	
DeptID	int	YES		NULL	
Email	varchar(100)	YES		NULL	

Result 7

Output

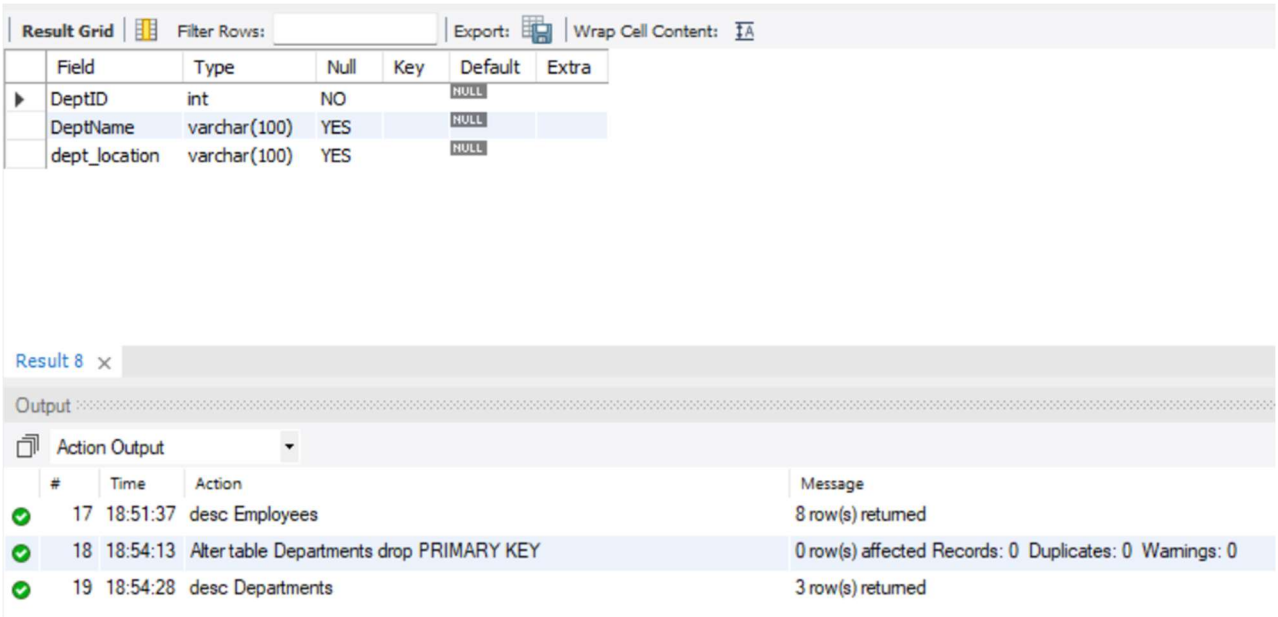
Action Output

#	Time	Action	Message
15	18:47:48	desc Employees	8 row(s) returned
16	18:51:21	Alter table Employees modify EmpName VARCHAR(120)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
17	18:51:37	desc Employees	8 row(s) returned

8. Drop the primary key Constraint from Departments

   Alter table Departments drop PRIMARY KEY;

   desc Departments;



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
DeptID	int	NO		NULL	
DeptName	varchar(100)	YES		NULL	
dept_location	varchar(100)	YES		NULL	

Result 8

Output

Action Output

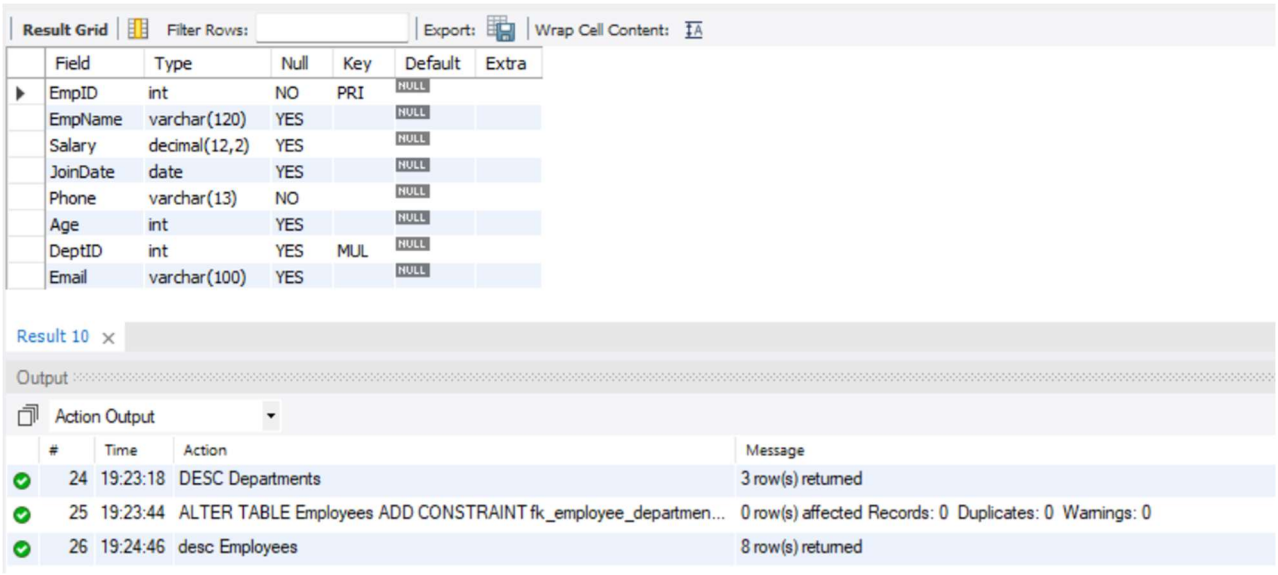
#	Time	Action	Message
17	18:51:37	desc Employees	8 row(s) returned
18	18:54:13	Alter table Departments drop PRIMARY KEY	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
19	18:54:28	desc Departments	3 row(s) returned

9. Create a foreign key constraint between two tables

   ALTER TABLE Employees ADD CONSTRAINT fk_employee_department

   FOREIGN KEY (DeptID) REFERENCES Departments(DeptID);

desc Employees;



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(120)	YES		NULL	
Salary	decimal(12,2)	YES		NULL	
JoinDate	date	YES		NULL	
Phone	varchar(13)	NO		NULL	
Age	int	YES		NULL	
DeptID	int	YES	MUL	NULL	
Email	varchar(100)	YES		NULL	

Result 10

Output

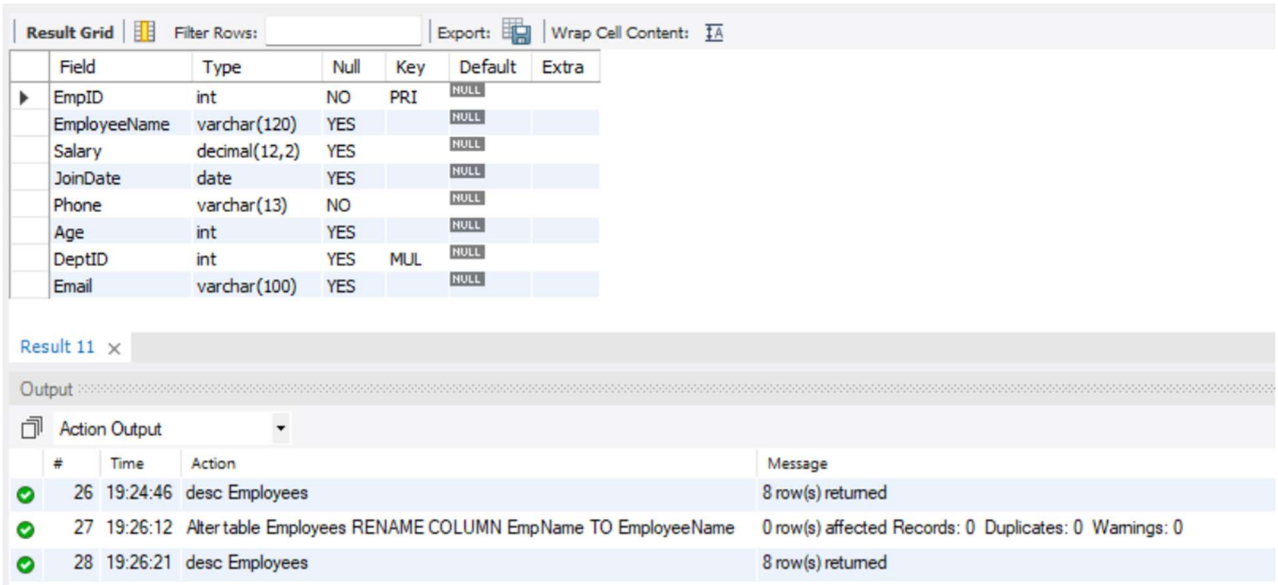
Action Output

#	Time	Action	Message
24	19:23:18	DESC Departments	3 row(s) returned
25	19:23:44	ALTER TABLE Employees ADD CONSTRAINT fk_employee_departmen...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
26	19:24:46	desc Employees	8 row(s) returned

10. Rename the column EmpName to EmployeeName.

Alter table Employees RENAME COLUMN EmpName TO EmployeeName;

desc Employees;



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmployeeName	varchar(120)	YES		NULL	
Salary	decimal(12,2)	YES		NULL	
JoinDate	date	YES		NULL	
Phone	varchar(13)	NO		NULL	
Age	int	YES		NULL	
DeptID	int	YES	MUL	NULL	
Email	varchar(100)	YES		NULL	

Result 11

Output

Action Output

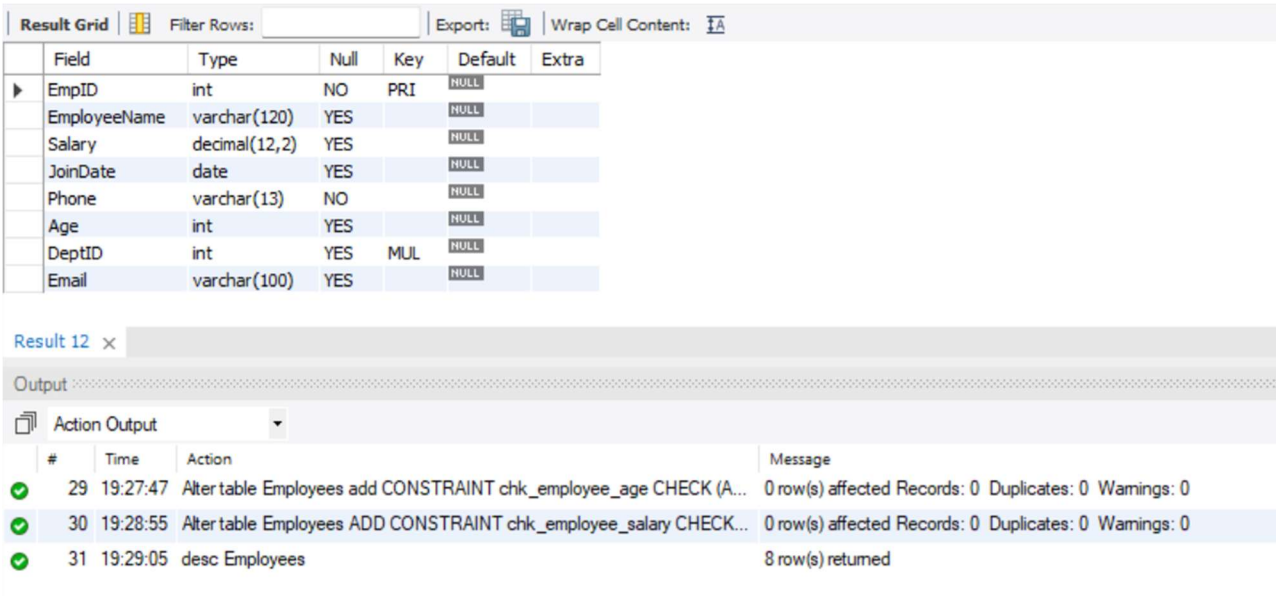
#	Time	Action	Message
26	19:24:46	desc Employees	8 row(s) returned
27	19:26:12	Alter table Employees RENAME COLUMN EmpName TO EmployeeName	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
28	19:26:21	desc Employees	8 row(s) returned

11. Apply a CHECK constraint on the Age and Salary columns.

Alter table Employees add CONSTRAINT chk_employee_age CHECK (Age >= 18);

Alter table Employees ADD CONSTRAINT chk_employee_salary CHECK (Salary >= 0);

desc Employees;



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmployeeName	varchar(120)	YES		NULL	
Salary	decimal(12,2)	YES		NULL	
JoinDate	date	YES		NULL	
Phone	varchar(13)	NO		NULL	
Age	int	YES		NULL	
DeptID	int	YES	MUL	NULL	
Email	varchar(100)	YES		NULL	

Result 12

Output

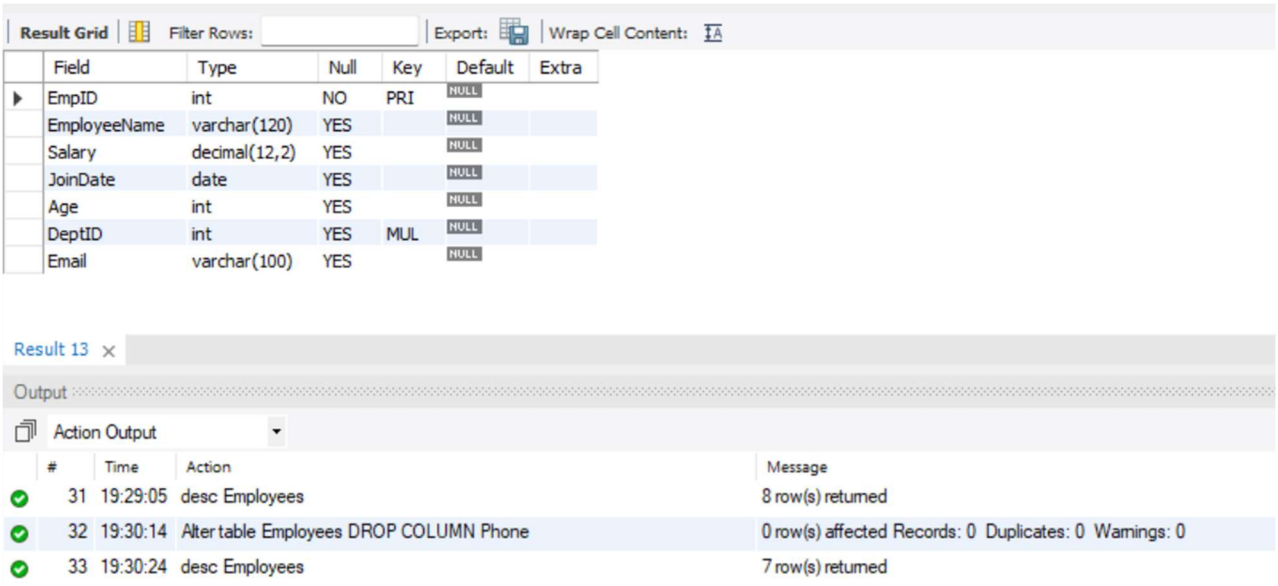
Action Output

#	Time	Action	Message
29	19:27:47	Alter table Employees add CONSTRAINT chk_employee_age CHECK (A...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
30	19:28:55	Alter table Employees ADD CONSTRAINT chk_employee_salary CHECK...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
31	19:29:05	desc Employees	8 row(s) returned

12. Remove the Phone column from the Employees table.

Alter table Employees DROP COLUMN Phone;

desc Employees;



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmployeeName	varchar(120)	YES		NULL	
Salary	decimal(12,2)	YES		NULL	
JoinDate	date	YES		NULL	
Age	int	YES		NULL	
DeptID	int	YES	MUL	NULL	
Email	varchar(100)	YES		NULL	

Result 13

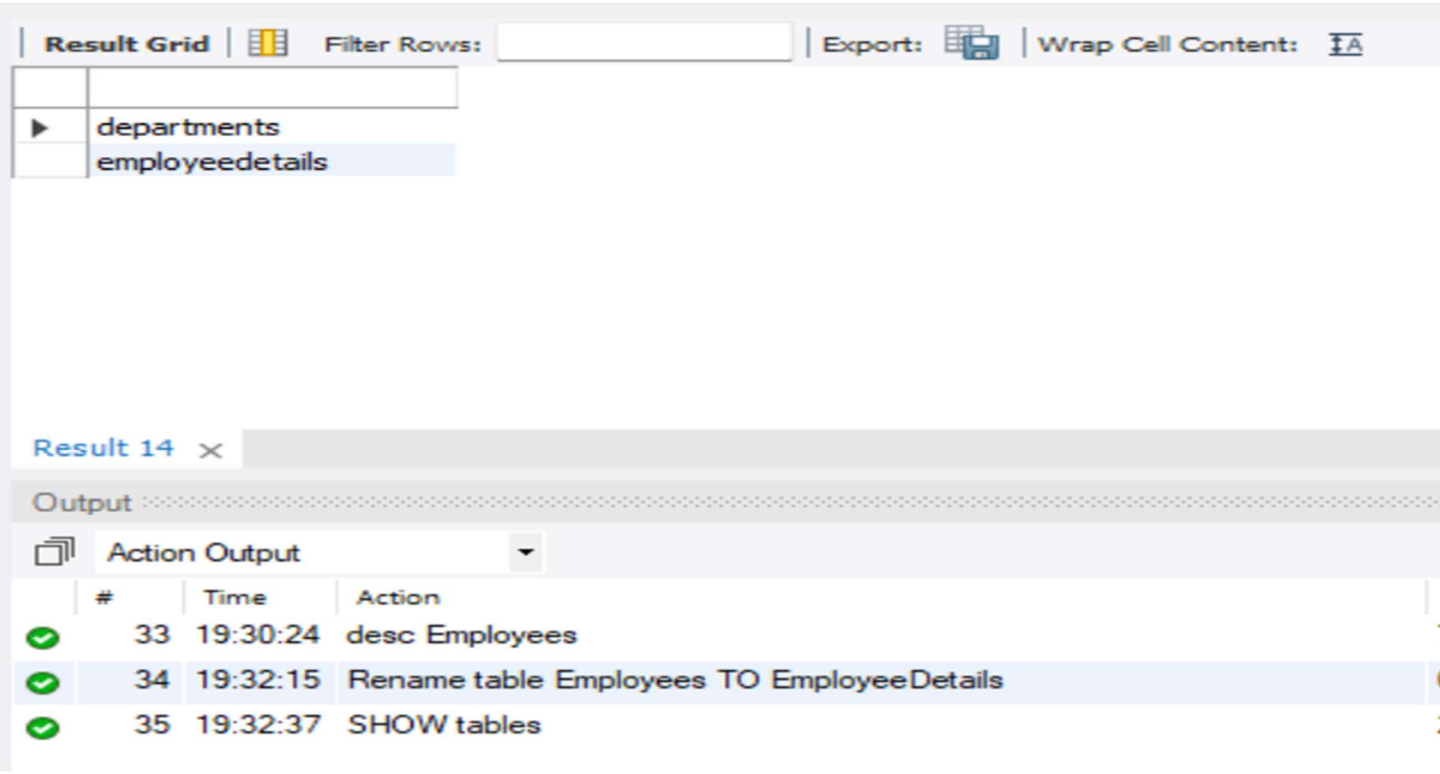
Output

Action Output

#	Time	Action	Message
31	19:29:05	desc Employees	8 row(s) returned
32	19:30:14	Alter table Employees DROP COLUMN Phone	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
33	19:30:24	desc Employees	7 row(s) returned

13. Rename the table Employees to EmployeeDetails.

```
Rename table Employees TO EmployeeDetails;
SHOW tables;
```



14. Drop the CHECK constraints.

```
Alter table  EmployeeDetails DROP CHECK chk_employee_age;
Alter table  EmployeeDetails DROP CHECK chk_employee_salary;
desc EmployeeDetails;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmployeeName	varchar(120)	YES		NULL	
Salary	decimal(12,2)	YES		NULL	
JoinDate	date	YES		NULL	
Age	int	YES		NULL	
DeptID	int	YES	MUL	NULL	
Email	varchar(100)	YES		NULL	

Result 15

Output

Action Output

#	Time	Action	Message
36	19:34:22	Alter table EmployeeDetails DROP CHECK chk_employee_age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
37	19:35:30	Alter table EmployeeDetails DROP CHECK chk_employee_salary	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
38	19:35:57	desc EmployeeDetails	7 row(s) returned

15. Delete the Departments table permanently.

    Drop table Departments;

    Show tables;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Tables_in_companydb
employeedetails

Result 16

Output

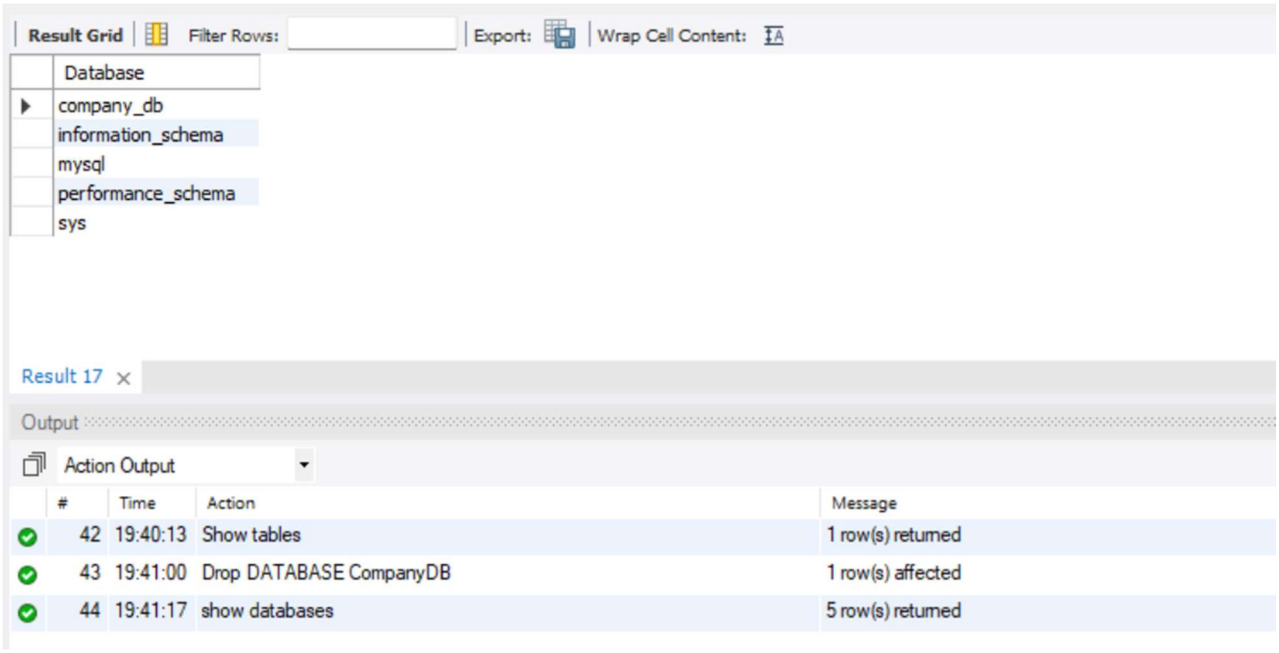
Action Output

#	Time	Action	Message
40	19:39:28	ALTER TABLE EmployeeDetails DROP FOREIGN KEY fk_employee_de...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
41	19:39:52	Drop table Departments	0 row(s) affected
42	19:40:13	Show tables	1 row(s) returned

16. Drop the database.

    Drop DATABASE CompanyDB;

    show databases;



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Database
company_db
information_schema
mysql
performance_schema
sys

Result 17

Output

Action Output

#	Time	Action	Message
42	19:40:13	Show tables	1 row(s) returned
43	19:41:00	Drop DATABASE CompanyDB	1 row(s) affected
44	19:41:17	show databases	5 row(s) returned